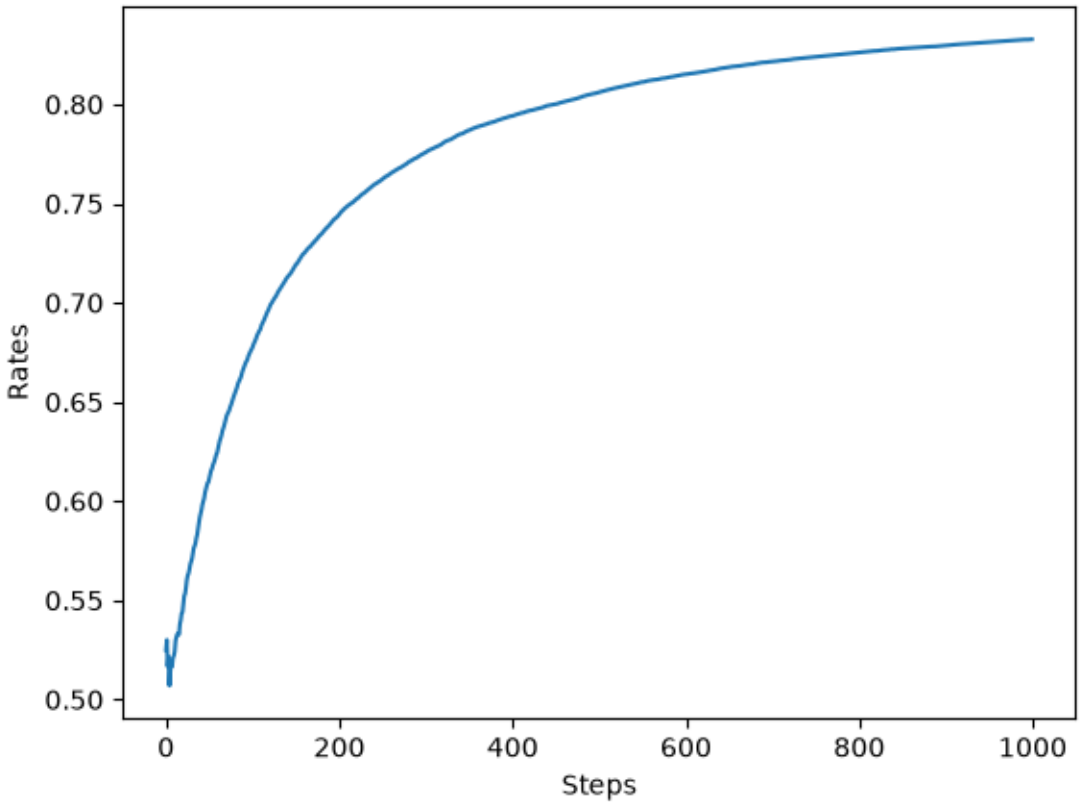
bandit_avg

September 14, 2026

[1]:
```
import numpy as np
import matplotlib.pyplot as plt
from bandit import Bandit, Agent


runs = 200
steps = 1000
epsilon = 0.1
all_rates = np.zeros((runs, steps))  # (200, 1000)

for run in range(runs):  # 200
    bandit = Bandit()
    agent = Agent(epsilon)
    total_reward = 0
    rates = []

    for step in range(steps):
        action = agent.get_action()
        reward = bandit.play(action)
        agent.update(action, reward)
        total_reward += reward
        rates.append(total_reward / (step + 1))

    all_rates[run] = rates  #

avg_rates = np.average(all_rates, axis=0)  #

# [  1-16]     (200      )
plt.ylabel('Rates')
plt.xlabel('Steps')
plt.plot(avg_rates)
plt.show()
```


0.80
0.75
0.70
0.65
0.60
0.55
0.50
Rates
0
200
400
600
800
1000
Steps